

PANEL 1/14 ? MASTER FRAME: GEOGRAPHY WITHOUT DETERMINISM

RELIEF + CLIMATE + WATER + SOIL + BIOTA + RESOURCES

-> opportunities, constraints and hazards

-> TECHNOLOGY + LABOUR + KNOWLEDGE + INSTITUTIONS + POWER

-> subsistence -> settlement -> exchange -> urbanisation -> polity

RULE: geography conditions; people select and transform possibilities.

HIMALAYAS / PASSES

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INDUS--THAR--GANGA-BRAHMAPUTRA

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VINDHYA-SATPURA / NARMADA-TAPI

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DECCAN--GHATS--COASTS/DELTA--ISLAND/STRAIT ZONES

Macro-regions contain micro-regions; modern borders are poor substitutes.

PANEL 3/14 ? HIMALAYAS, PASSES AND FRONTIERS

BARRIER: climate modification | defence cost | rugged relief

CORRIDOR: valleys | Khyber-Bolan approaches | seasonal movement

FLOWS: trade | pastoralism | migration | texts | ideas | armies

Frontier = zone with seasonally and politically variable permeability.

TRAP: an invasion route is not the only history of a pass.

PANEL 4/14 ? RIVERS, BASINS AND WATERSHEDS

INDUS/GHAGGAR-HAKRA: floodplain + semi-arid adaptation + channel change

GANGA-BRAHMAPUTRA: alluvium + wetlands + transport + flood/erosion

PENINSULAR: Narmada-Tapi corridors; east-flowing rivers to deltas/coasts

BASIN collects drainage; WATERSHED DIVIDE separates adjacent basins.

Rivers = water + route + risk + memory, never irrigation alone.

PALAEOCHANNEL -> geomorphology and dated flow history

SETTLEMENT CLUSTER -> archaeological occupation and ecology

SARASWATI -> textual name, layer, genre and remembered geography

Link only after independent chronology and scale comparison.

TRAP: convergence can support a hypothesis; it cannot erase uncertainty.

PANEL 6/14 ? MONSOON, SOILS, FORESTS AND GRAZING

SW MONSOON | NE MONSOON | WINTER RAIN -> different calendars

ALLUVIUM | BLACK | RED/LATERITIC SOILS -> different crop-water choices

FORESTS -> communities + fuel + timber + elephants + pasture + frontier

UNCERTAINTY -> storage + mobility + crop diversity + water institutions

TRAP: one dry proxy is not an all-India collapse narrative.

PANEL 7/14 ? RESOURCE CHAINS, NOT RESOURCE DETERMINISM

DEPOSIT -> extraction -> skill/fuel/labour -> route -> demand/control

Khetri copper | Rohri chert | eastern/central iron | scarce tin

salt/shell/fish/pearls | timber | elephants | building stone

A resource map is not a production, trade or political-control map.

Add ecological cost: mining and smelting can intensify fuel demand.

PANEL 8/14 ? SETTLEMENT ECOLOGY

SITE: water + soil + pasture + drainage + defence + raw material + route

FUNCTION: camp -> village -> workshop -> town -> port -> capital

NETWORK: hinterland supplies food + fuel + labour + craft materials

Mohenjo-daro floodplain | Dholavira reservoirs | Lothal coastal interface

TRAP: river proximity does not create one river-civilisation formula.

ANCIENT HISTORY 03

PANEL 9/14 ? CHRONOLOGY: ASYNCHRONOUS BUT INTERCONNECTED

PREHISTORY: mobile use of water/raw-stone landscapes

FOOD PRODUCTION: regional crops/herds; Koldihwa-Mahagara debate

CHALCOLITHIC: stone-copper overlap; farming/herding mixes

HARAPPAN -> VEDIC/POST-VEDIC -> EARLY HISTORIC regional integration

No all-India Stone -> Copper -> Iron timetable.

PANEL 10/14 ? HARAPPAN ECOLOGICAL MOSAIC AND RESILIENCE

river plains + piedmonts + deserts + semi-arid zones + coasts

Mohenjo-daro | Dholavira | Lothal | Kalibangan = varied adaptations

hydrological stress may matter; social/economic chronology also matters

late Harappan dispersal includes continuity, relocation and reorganisation

TRAP: transformation is safer than monocausal disappearance.

PANEL 11/14 ? GANGA PLAINS, IRON, WET RICE AND STATE FORMATION

alluvium + rainfall + river routes = potential

iron tools + crop regimes + labour + clearance = production change

surplus + taxation + warfare + capitals + institutions = state capacity

Atranjikhhera | Rajgir | Pataliputra = named evidence

TRAP: neither iron nor wet rice alone caused Magadha.

central forests/minerals + passes + Narmada-Tapi corridors

Deccan soils/plateaus + mixed farming/pastoralism + regional cultures

east-flowing rivers -> Ghats openings -> deltas -> port hinterlands

Arikamedu imports show contact; islands/straits form distinct ecologies

TRAP: ports need inland production, vessels, merchants and security.

PANEL 13/14 ? ENVIRONMENTAL CHANGE AND PROXY LIMITS

pollen/seeds/fauna/sediments/isotopes/palaeochannels/GIS

OBSERVATION -> date/scale -> bounded inference -> corroboration

EQUIFINALITY: drought/flood/conflict/disease/route change can look alike

RESILIENCE: mobility + storage + crops + herding + exchange + institutions

A proxy constrains context; it does not narrate society by itself.

- 1 Decode directive and period.
 - 2 Label only sites/rivers/routes used in the argument.
 - 3 Write factor -> named evidence -> mechanism -> outcome.
 - 4 Add technology/labour/institution and source or scale limit.
 - 5 Conclude: regionally uneven choices; reciprocal landscape change.
- 2023 GS-I: explain role, do not submit a geographical catalogue.